LAB 02 - WRITING MYSQL QUERIES 1
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1.Recreate the following tables with relationships using MySQL syntaxes.

```
mysql>  CREATE TABLE Publisher(
    ->     PubID varchar(6) PRIMARY KEY,
    ->     Publishername  varchar(30),
    ->     RegisteredDate date,
    ->     Country varchar(15)
    ->     );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC Publisher;
+----------------+-------------+------+-----+---------+-------+
| Field          | Type        | Null | Key | Default | Extra |
+----------------+-------------+------+-----+---------+-------+
| PubID          | varchar(6)  | NO   | PRI | NULL    |       |
| Publishername  | varchar(30) | YES  |     | NULL    |       |
| RegisteredDate | date        | YES  |     | NULL    |       |
| Country        | varchar(15) | YES  |     | NULL    |       |
+----------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Book(
    ->    BookID varchar(5) PRIMARY KEY,
    ->    Title varchar(50),
    ->    Author varchar(30),
    ->    Category int,
    ->    Price float,
    ->    PubID varchar(6),
    ->    FOREIGN KEY(PubID) REFERENCES Publisher(PubID)
    ->    );
Query OK, 0 rows affected (0.08 sec)

mysql> DESC Book;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| BookID   | varchar(5)  | NO   | PRI | NULL    |       |
| Title    | varchar(50) | YES  |     | NULL    |       |
| Author   | varchar(30) | YES  |     | NULL    |       |
| Category | int         | YES  |     | NULL    |       |
| Price    | float       | YES  |     | NULL    |       |
| PubID    | varchar(6)  | YES  | MUL | NULL    |       |
+----------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

2. Insert following records to Publisher table and Book.

```
mysql>  INSERT INTO Publisher(PubID,PublisherName,RegisteredDate,Country)
    ->     VALUES
    ->     ('Pub01','A.Press','2001-05-12','Sri Lanka'),
    ->     ('Pub02','A.Books','2010-03-20','India'),
    ->     ('Pub03','K.Brill','2014-01-31','Poland'),
    ->     ('Pub04','A.Press','2015-02-13','Sri Lanka');
Query OK, 4 rows affected (0.02 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Book(BookID,Title,Author,Category,Price,PubID)
    ->     VALUES
    ->     ('B1','Excel','R.David',2,18.00,'Pub02'),
    ->     ('B2','Computers','M.Thomas',4,25.00,'Pub02'),
    ->     ('B3','Access','P.Paul',2,30.00,'Pub03'),
    ->     ('B4','Arts','W.Shiva',1,10.00,'Pub04'),
    ->     ('B5','Science','A.Rahul',3,9.00,'Pub01')
    ->     ;
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

RESULTS,

```
mysql> SELECT * from Book;
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B1     | Excel     | R.David  |        2 |    18 | Pub02 |
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
| B3     | Access    | P.Paul   |        2 |    30 | Pub03 |
| B4     | Arts      | W.Shiva  |        1 |    10 | Pub04 |
| B5     | Science   | A.Rahul  |        3 |     9 | Pub01 |
+--------+-----------+----------+----------+-------+-------+
5 rows in set (0.01 sec)

mysql> SELECT * from Publisher;
+-------+---------------+----------------+-----------+
| PubID | Publishername | RegisteredDate | Country   |
+-------+---------------+----------------+-----------+
| Pub01 | A.Press       | 2001-05-12     | Sri Lanka |
| Pub02 | A.Books       | 2010-03-20     | India     |
| Pub03 | K.Brill       | 2014-01-31     | Poland    |
| Pub04 | A.Press       | 2015-02-13     | Sri Lanka |
+-------+---------------+----------------+-----------+
4 rows in set (0.00 sec)
```

3. Write SQL queries/statements considering the following table. Table Name: Book

a. Display all the details in the Book table.

```
mysql> SELECT * FROM Book;
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B1     | Excel     | R.David  |        2 |    18 | Pub02 |
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
| B3     | Access    | P.Paul   |        2 |    30 | Pub03 |
| B4     | Arts      | W.Shiva  |        1 |    10 | Pub04 |
| B5     | Science   | A.Rahul  |        3 |     9 | Pub01 |
+--------+-----------+----------+----------+-------+-------+
5 rows in set (0.00 sec)
```

B. Display all the titles of the books.

```
mysql> SELECT Title FROM Book;
+-----------+
| Title     |
+-----------+
| Excel     |
| Computers |
| Access    |
| Arts      |
| Science   |
+-----------+
5 rows in set (0.00 sec)
```

C. Display all the titles of the books with prices.

```
mysql> SELECT Title,Price FROM Book;
+-----------+-------+
| Title     | Price |
+-----------+-------+
| Excel     |    18 |
| Computers |    25 |
| Access    |    30 |
| Arts      |    10 |
| Science   |     9 |
+-----------+-------+
5 rows in set (0.00 sec)
```

D. Display the Book ID, the Title and the Author.

```
mysql> SELECT BookID,Title,Author FROM BOOK;
+--------+-----------+----------+
| BookID | Title     | Author   |
+--------+-----------+----------+
| B1     | Excel     | R.David  |
| B2     | Computers | M.Thomas |
| B3     | Access    | P.Paul   |
| B4     | Arts      | W.Shiva  |
| B5     | Science   | A.Rahul  |
+--------+-----------+----------+
5 rows in set (0.00 sec)
```

e. Display the details of the book with the Book ID B2.

```
mysql> SELECT * FROM Book WHERE BookID='B2';
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
+--------+-----------+----------+----------+-------+-------+
1 row in set (0.00 sec)
```

f. Display the titles of the books with the prices less than 20.00.

```
mysql> SELECT Title FROM Book WHERE Price<20;
+---------+
| Title   |
+---------+
| Excel   |
| Arts    |
| Science |
+---------+
3 rows in set (0.00 sec)
```

g. Display the details of the books in category 2.

```
mysql> SELECT * FROM Book WHERE Category=2;
+--------+--------+---------+----------+-------+-------+
| BookID | Title  | Author  | Category | Price | PubID |
+--------+--------+---------+----------+-------+-------+
| B1     | Excel  | R.David |        2 |    18 | Pub02 |
| B3     | Access | P.Paul  |        2 |    30 | Pub03 |
+--------+--------+---------+----------+-------+-------+
2 rows in set (0.00 sec)
```

4. Write queries

a. Display all the Titles in the Book table in ascending order.

```
mysql> SELECT Title FROM Book ORDER BY Title ASC;
+-----------+
| Title     |
+-----------+
| Access    |
| Arts      |
| Computers |
| Excel     |
| Science   |
+-----------+
5 rows in set (0.00 sec)
```

b. Display all the details in the Book table in descending order by Price.

```
mysql> SELECT * FROM Book ORDER BY Price DESC;
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B3     | Access    | P.Paul   |        2 |    30 | Pub03 |
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
| B1     | Excel     | R.David  |        2 |    18 | Pub02 |
| B4     | Arts      | W.Shiva  |        1 |    10 | Pub04 |
| B5     | Science   | A.Rahul  |        3 |     9 | Pub01 |
+--------+-----------+----------+----------+-------+-------+
5 rows in set (0.00 sec)
```

c. Display all the details in the Book table in ascending order by Price and Title.

```
mysql> SELECT * FROM Book ORDER BY Price,Title ASC;
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B5     | Science   | A.Rahul  |        3 |     9 | Pub01 |
| B4     | Arts      | W.Shiva  |        1 |    10 | Pub04 |
| B1     | Excel     | R.David  |        2 |    18 | Pub02 |
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
| B3     | Access    | P.Paul   |        2 |    30 | Pub03 |
+--------+-----------+----------+----------+-------+-------+
5 rows in set (0.00 sec)
```

5. Write SQL queries/statements considering the ‘Book’ table

a. Change the category as 3 of the book with the BookID ‘B3’.

```
mysql> UPDATE Book SET Category=3 WHERE BookID='B3';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECt * from Book WHERE BookID='B3';
+--------+--------+--------+----------+-------+-------+
| BookID | Title  | Author | Category | Price | PubID |
+--------+--------+--------+----------+-------+-------+
| B3     | Access | P.Paul |        3 |    30 | Pub03 |
+--------+--------+--------+----------+-------+-------+
1 row in set (0.00 sec)
```

b. Change the price of the book with the BookID ‘B3’ by adding 10.00 to the current price.

```
mysql> UPDATE Book
    -> SET Price = Price + 10.00
    -> WHERE BookID = 'B3';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * from Book WHERE BookID='B3';
+--------+--------+--------+----------+-------+-------+
| BookID | Title  | Author | Category | Price | PubID |
+--------+--------+--------+----------+-------+-------+
| B3     | Access | P.Paul |        3 |    40 | Pub03 |
+--------+--------+--------+----------+-------+-------+
1 row in set (0.00 sec)
```

c. Delete the details of the book with the BookID ‘B3’.

```
mysql> DELETE from Book WHERE BookID='B3';
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * from Book;
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B1     | Excel     | R.David  |        2 |    18 | Pub02 |
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
| B4     | Arts      | W.Shiva  |        1 |    10 | Pub04 |
| B5     | Science   | A.Rahul  |        3 |     9 | Pub01 |
+--------+-----------+----------+----------+-------+-------+
4 rows in set (0.00 sec)
```

d. Delete all the books in category 2.

```
mysql> DELETE from Book WHERE Category=2;
Query OK, 1 row affected (0.03 sec)

mysql> SELECT * from Book;
+--------+-----------+----------+----------+-------+-------+
| BookID | Title     | Author   | Category | Price | PubID |
+--------+-----------+----------+----------+-------+-------+
| B2     | Computers | M.Thomas |        4 |    25 | Pub02 |
| B4     | Arts      | W.Shiva  |        1 |    10 | Pub04 |
| B5     | Science   | A.Rahul  |        3 |     9 | Pub01 |
+--------+-----------+----------+----------+-------+-------+
3 rows in set (0.00 sec)
```

e. Remove all the data in the Book table.

```
mysql> DELETE from Book;
Query OK, 3 rows affected (0.01 sec)

mysql> SELECT * from Book;
Empty set (0.00 sec)
```